

Jadu Multimodal AI Engineer Challenge

Context

At Jadu, creatives ask us to generate scenes, shots, and short sequences. These requests can be realistic or fantastical.

We generate **images and short videos** using prompts and, in some cases, **reference images**. A request may go through multiple steps: selecting a model, preparing or rewriting a prompt, generating an asset, running quality control on the result, and sometimes retrying with a revised prompt or a different model.

The linked files ([qc_results.json](#) and [qc_results_video.json](#)) contain examples from that workflow. Depending on the entry, they may include:

- prompts
- model metadata
- reference or input image URLs
- generated output URLs
- QC scores
- QC reasoning / actionable feedback
- and, in some video cases, a QA-rewritten prompt

For this exercise, assume these files are the **only data available** to you.

We want to understand how you think about:

- improving prompts and system prompts
- building trustworthy evaluation
- preparing datasets / eval sets / regression sets
- defining useful metrics
- improving the workflow using evidence from real examples

The current QC should **not** be assumed to be perfect. Part of the exercise is to assess where it is useful, where it is weak, and how you would improve it.

0) Relevant hands-on experience (required)

Before your proposal, describe up to **2 systems or projects** you personally worked on that are relevant to this role.

We are especially interested in hands-on experience with:

- improving **prompts or system prompts**
- designing or calibrating **automated evaluation**
- preparing **datasets / eval sets / regression sets**
- choosing and interpreting **metrics**
- improving workflow quality through iteration and experiments

For each example, please cover:

Context

What was the product or workflow?

Your role

What did you personally own or implement?

Prompting / system prompting

Describe one concrete prompt or system prompt change you made.

Why did you make it? What failure mode were you trying to fix?

Evaluation

Describe the evaluation setup.

For example: human review, pairwise comparison, rubric-based QA, LLM-as-judge, VLM-based evaluation, regression suite, or another method.

Dataset preparation

Explain how you prepared the dataset or eval set.

How did you choose examples, slices, hard cases, or failure buckets?

Metrics

Which metrics did you use?

Which ones were most useful, and which ones were misleading?

Outcome

How did you know the change was better?

Did you observe regressions or tradeoffs?

Reflection

What would you do differently today?

Please be explicit about what was **hands-on work you personally did** versus what is a **proposal or hypothesis**.

If there is an area you have **not** done hands-on, say so clearly. We value honesty much more than broad generic claims.

1) Audit the data and the current QA

Review the two dumps and tell us:

- what data you would trust
- what data you would treat carefully
- what data you would exclude from analysis or production decisions
- what issues in the logs or QA process could affect system design

Ground this section in specific evidence.

Please reference **some joblds** across the two dumps, including at least:

- **3 image jobs**
- **3 video jobs**

For each cited example, briefly say:

- whether you agree with the existing QA
- what the main failure mode is, if any
- what the system should have done next
(accept, retry, prompt change, switch model, human review, ask for clarification, etc.)

Please include **at least 2 examples** where you believe the current QA is wrong, incomplete, or not useful enough for decision-making.

We care much more about specific judgment tied to concrete examples than about broad general statements.

2) Design V2 of the workflow

Using only the supplied data, explain how you would improve the pipeline for both **image** and **video** generation.

Please cover:

- how you would choose a model
- how you would write or rewrite prompts
- how you would evaluate whether an output is good enough
- how you would decide whether to accept, retry, switch model, or escalate
- where you would use automated judges, where you would not, and why

Your answer must include:

- **one example of a revised system prompt or evaluation prompt**
- **one proposed eval set design**, including slices or buckets you would track
- **one regression suite proposal**
- **the metrics and thresholds** you would use for release decisions
- **where automated evaluation is allowed to decide alone**, and where it must defer to human review

To make your proposal concrete, walk through **2 example creative requests of your choice**. At least one can be fantastical.

For each example, show:

- model choice
- prompt strategy
- acceptance rule
- fallback / retry path

Optional: if you have relevant experience, briefly explain how you would extend the same approach to **audio generation**.

Please also mention any important production constraints you would keep in mind (for example latency, retry strategy, or cost), but keep the focus on workflow quality and evaluation.

3) Measure improvement and prevent regressions

Assume we are releasing **V2** of the pipeline.

Using only the supplied dumps, explain:

- how you would define success
- what your **primary quality metrics** would be
- what your **secondary / guardrail metrics** would be
- how you would define **pass / fail thresholds**
- how you would create **evaluation slices**
(for example by prompt type, complexity, reference-image usage, motion difficulty, character consistency, etc.)
- how you would validate or calibrate automated QA
- how you would detect when the automated evaluator is drifting or overconfident
- what would count as a regression
- what would block release even if the aggregate metric improved

Please include at least:

- **1 example of a metric you would trust**
- **1 example of a metric you would treat carefully**
- **1 example of a failure case that should become a permanent regression test**

We care about practical release quality, not just offline averages.

Deliverables

Please submit:

- a short memo; **2 pages is ideal, 4 pages max**
- a small appendix or table with the **joblds** you reviewed and what you concluded from each
- optional: code, notebook, or diagrams

Concise, well-supported reasoning is much more valuable than a long document.

Use of GenAI tools

You may use GenAI tools.

If you do, include a short note at the end covering:

- what you used them for

- what parts reflect your own judgment
- where you agreed or disagreed with the tool output

For every major recommendation, please label it as either:

- **based on hands-on experience**, or
 - **proposal / hypothesis for this exercise**
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What we value

We are looking for:

- strong judgment grounded in actual examples
- clear understanding of multimodal workflows in practice
- good taste around evaluation and automated QA
- thoughtful prompt / system prompt reasoning
- practical understanding of datasets, eval sets, regression sets, and metrics
- concise, specific reasoning
- honesty about what you have done hands-on versus what you would propose

Generic answers that are not clearly grounded in the provided logs will score poorly.